



UNIVERSITY OF
BIRMINGHAM

Superconductivity

Lecture 09 – “Fields and Currents”

Dr Mingee Chung (Minki Jeong)

Associate Professor

Condensed Matter Physics Group

School of Physics & Astronomy

Previously, Ginzburg-Landau equation & coherence length

Ginzburg-Landau (GL) equation for superconductors with spatial inhomogeneity:

$$-\frac{\hbar^2}{2m^*}\nabla^2\psi + a\psi + b\psi|\psi|^2 = 0, \quad (1)$$

Solution for metal-superconductor interface:

$$\psi(x) = \psi_\infty \tanh\left(\frac{x}{\sqrt{2}\xi(T)}\right). \quad (2)$$

Ginzburg-Landau coherence length:

$$\xi(T) \equiv \left(\frac{\hbar^2}{2m^*|a(T)|}\right)^{1/2}, \quad (3)$$

T dependence of the GL coherence length:

$$\xi(T) = \xi(0)|t|^{-1/2} \quad (4)$$

Table of Contents

Magnetic vector potential

Critical field

Critical current

Magnetic vector potential

Penetration depth

By introducing a magnetic field, one can define another length scale.
Guess what? *penetration depth* λ .

For instance, by combining

- $\lambda = \sqrt{m^*/\mu_0 n_s e^{*2}}$ (from London equation),
- $|\psi|^2 = n_s = -a/b$,

one obtains

$$\lambda(T) = \sqrt{\frac{m^* b}{\mu_0 e^{*2} |a(T)|}}. \quad (5)$$

Magnetic vector potential

Follow the standard procedure in quantum mechanics,

$$-i\hbar\nabla \Rightarrow -i\hbar\nabla - q\mathbf{A}, \quad (6)$$

where we may set $q = -e^*$, such that

$$f_s - f_n = a|\psi(\mathbf{r})|^2 + \frac{b}{2}|\psi(\mathbf{r})|^4 - \frac{\hbar^2}{2m^*}|\nabla\psi(\mathbf{r})|^2, \quad (7)$$

becomes

$$f_s - f_n = a|\psi(\mathbf{r})|^2 + \frac{b}{2}|\psi(\mathbf{r})|^4 + \frac{1}{2m^*}|(-i\hbar\nabla + e^*\mathbf{A})\psi(\mathbf{r})|^2, \quad (8)$$

One can repeat the functional derivative of the total free energy w.r.t. ψ and ψ^* to obtain a full GL equation

$$\frac{1}{2m^*}(-i\hbar\nabla + e^*\mathbf{A})^2\psi + a\psi + b\psi|\psi|^2 = 0. \quad (9)$$

Ginzburg-Landau equation

Solving

$$\frac{1}{2m^*} (-i\hbar\nabla + e^*\mathbf{A})^2 \psi + a\psi + b\psi|\psi|^2 = 0. \quad (10)$$

is challenging in general. We'll do it under certain assumption in the next lecture.

In the remainder, we'll step back to find the free energy minimum expression for simple cases (e.g., thin slab or wire) to obtain the critical field and current.

Critical field

A thin slab in a parallel field

What's your expectation for a critical field if a slab gets thinner?

⇒ The critical field is increased from the bulk value, since larger penetration of the field reduces energy cost due to diamagnetism.

Let say x refers to the direction normal to the slab of thickness d (a field applied along z direction).

One can use the Landau gauge $\mathbf{A} = (0, \mu_0 H_a x, 0)$,
since $\mathbf{B} = \nabla \times \mathbf{A} = \mu_0 H_a \hat{z}$.

$$F_s = \int_{-d/2}^{d/2} f_s dx \quad (11)$$

$$= \int_{-d/2}^{d/2} \left[f_n + a|\psi|^2 + \frac{b}{2}|\psi|^4 + \frac{1}{2m^*} |(e^* \mu_0 H x) \psi|^2 \right] dx, \quad (12)$$

which you can easily evaluate.

Critical field reduced

Now, by minimizing F_s with respect to $|\psi|^2$,

$$\frac{\partial F_s}{\partial |\psi|^2} = \left(ad + \frac{e^{*2} H^2 d^3}{24 m^*} \right) + bd |\psi|^2 \quad (13)$$

$$= 0, \quad (14)$$

one would get

$$|\psi|^2 = |\psi_\infty|^2 \left[1 - \frac{1}{24} \left(\frac{d}{\lambda} \right)^2 \left(\frac{H}{H_c} \right)^2 \right]. \quad (15)$$

This allows us to find the new critical field by setting $|\psi|^2 = 0$, i.e.,

$$H'_c = 2\sqrt{6} H_c \left(\frac{\lambda}{d} \right). \quad (16)$$

Critical current

A thin wire (1/4)

Suppose a super-current

$$\mathbf{j}_s = e^* |\psi|^2 \mathbf{v}_s. \quad (17)$$

We may further write down the free energy density difference as

$$f_s - f_n = a |\psi|^2 + \frac{1}{2} b |\psi|^4 + \frac{1}{2} m^* v_s^2 |\psi|^2. \quad (18)$$

By minimizing the above with respect to $|\psi|^2$,

$$\frac{\partial}{\partial |\psi|^2} (f_s - f_n) = a + b |\psi|^2 + \frac{1}{2} m^* v_s^2 \quad (19)$$

$$= 0, \quad (20)$$

one obtains

$$|\psi|^2 = -\frac{a}{b} \left(1 - \frac{m^* v_s^2}{2|a|} \right) \quad (21)$$

$$= |\psi_\infty|^2 \left(1 - \frac{m^* v_s^2}{2|a|} \right) = |\psi_\infty|^2 \left[1 - \left(\frac{\xi m^* v_s}{\hbar} \right)^2 \right]. \quad (22)$$

A thin wire (2/4)

Using

$$|\psi|^2 = |\psi_\infty|^2 \left[1 - \left(\frac{\xi m^* v_s}{\hbar} \right)^2 \right]. \quad (23)$$

the super-current density can be written as

$$\mathbf{j}_s = e^* |\psi|^2 \mathbf{v}_s \quad (24)$$

$$= e^* |\psi_\infty|^2 \left[1 - \left(\frac{\xi m^* v_s}{\hbar} \right)^2 \right] \mathbf{v}_s. \quad (25)$$

What does this mean?

- n_s decreases with increasing v_s ;
- there exists a maximum current with increasing v_s .

A thin wire (3/4)

$$\frac{\partial j_s}{\partial v_s} = e^* |\psi_\infty|^2 - 3e^* |\psi_\infty|^2 \left(\frac{\xi m^* v_s}{\hbar} \right)^2 \quad (26)$$

$$= e^* |\psi_\infty|^2 \left[1 - 3 \left(\frac{\xi m^* v_s}{\hbar} \right)^2 \right] \quad (27)$$

$$= 0. \quad (28)$$

From the above, the supercurrent velocity when the current is maximum becomes

$$v_s = \frac{1}{\sqrt{3}} \frac{\hbar}{\xi m^*}. \quad (29)$$

A thin wire (4/4)

The maximal current is obtained when $|\psi|^2 = (2/3)|\psi_\infty|^2$. Now, plugging this back to the expression for j_s gives the maximum value as

$$j_{s,\max} = \frac{2e^*\hbar n_s}{3\sqrt{3}m^*\xi} \propto (1 - T/T_c)^{3/2}, \quad (30)$$

which we may consider as the critical current density j_c .

Within the London, the critical current density is expressed as

$$j_c = H_c / \lambda. \quad (31)$$

which shows 1.84 times larger value than the GL one. Why?